CURRICULUM VITAE
DA-WU XIAO

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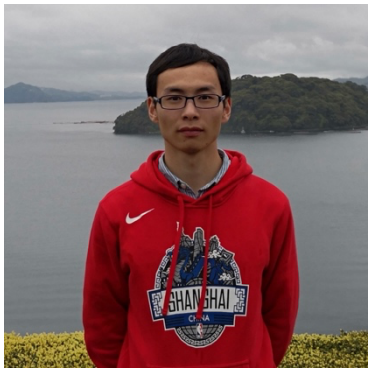

PERSONAL INFORMATION

Birth Year	May 1994
Birthplace	Shaanxi Ankang, China
Gender	Male
Nationality	China

EDUCATION

Sep 2016- Jul 2021, Ph.D. in Condensed Matter Physics (with Outstanding Graduates), Beijing Computational Science Research Center, China Academy of Engineering Physics, Beijing, China
**Thesis: Studies on the Spin Dynamics in a Few Quantum Systems*

Sep 2012- Jul 2016, BSc in Physics, School of Physical Science and Technology, Lanzhou University, Gansu, China

EMPLOYMENT

Nov 2021– present, Postdoctoral Fellow, Department of Physics, The Chinese University of Hong Kong, Hong Kong SAR

RESEARCH INTERESTS

- Quantum optics interplay with many-body spin system
- Spin dynamics and open quantum system
- Quantum technologies using spins and light

PUBLICATIONS

[1] **D.-W. Xiao**, C. Chen, and R. B. Liu, "Squeezed superradiant lasing of a quantum many-body emitter", arXiv:2602.16215 (2026).

[2] **D.-W. Xiao**, and C. Chen, "Metastability in the dissipative quantum Rabi model", Physical Review A **113**, 013735 (2026).

[3] X. Zhang, J. Hu, **D.-W. Xiao**, and N. Zhao, "Frequency shift caused by nonuniform field and boundary relaxation in magnetic resonance and comagnetometers", Commun. Phys. **8**, 93 (2025).

[4] **D.-W. Xiao**, C. Chen, and R.B. Liu, "Spontaneously squeezing superradiant laser", CLEO: Fundamental Science, FW4J.7 (2024).

[5] Y. Chen, **D.-W. Xiao**, and P. Zhang, "Three ultracold fermions in a two-dimensional anisotropic harmonic confinement", Commun. Theor. Phys. **74**, 045501 (2022).

[6] **D.-W. Xiao**, R. Zhang, and P. Zhang, "Universal energy-dependent pseudopotential for two-body problem of confined ultracold atoms", Phys. Rev. Research **4**, 013112 (2022).

[7] T. Kornher, **D.-W. Xiao**, K. Xia, F. Sardi, N. Zhao, R. Kolesov, and J. Wrachtrup, "Sensing Individual Nuclear Spins with a Single Rare-Earth Electron Spin", Phys. Rev. Lett. **124**, 170402 (2020).

[8] Y. Chen, **D.-W. Xiao**, R. Zhang, and P. Zhang, "Analytical solution for the spectrum of two ultracold atoms in a completely anisotropic confinement", Phys. Rev. A **101**, 053624 (2020).

[9] **D.-W. Xiao**, W.-H. Hu, Y. Cai and N. Zhao, "Magnetic Noise Enabled Biocompass", Phys. Rev. Lett. **124**, 128101 (2020).

[10] **D.-W. Xiao**, R. Zhang, and P. Zhang, "Confinement Induced Resonance with Weak Bare Interaction in a Quasi 3+0 Dimensional Ultracold Gas", Few-Body Systems **60**, 4 (2019).

[11] Z. Zhu, Y. Song, X. Liu, **D.-W. Xiao**, L. Yang, and L. Huang, "Observation of alternately localized Faraday waves in a narrow tank", Phys. Rev. Fluids **4**, 014807 (2019).

ACTIVITIES AND TALKS

[1] Apr 2021, Jiangsu, China. The First Applied Physics Forum by IOP. Talk: Magnetic noise enabled biocompass.

[2] May 2024, Charlotte, USA. CLEO 2024: Conference on Lasers and Electro-Optics. Talk: Spontaneously squeezing superradiant laser.

[3] Aug 2024, Hong Kong, China. The 2024 Hong Kong Summer Workshop on Spin-based Quantum Science and Technology. Talk: Lasing from an interacting spin system.

[4] Dec 2024, Okinawa, Japan. Impurity Spins for Quantum Information Science and Technologies (iSQIT24). Talk: Superradiant lasing from an interacting spin system.

[5] Aug 2025, Hong Kong, China. The 2025 Hong Kong Summer Workshop on Spin-based Quantum Science and Technology. Talk: Squeezed superradiant lasing of a quantum many-body emitter.